

Head Position Processing in the MEG Preprocessing Pipeline

(Centering, cHPI Pose Estimation, Movement Compensation, and YAML Configuration)

1. The Overall Goal

Head-position processing in MEG has **two distinct purposes**, which are often conflated:

1. Estimating the subject's pose over time

- derived from **continuous HPI coils (cHPI)**.
- yields time-varying translations + rotations of the head relative to the sensors.
- used **only** for *movement compensation* in Maxwell filtering.

2. Estimating the head center (SSS "origin") in the HEAD frame

- derived from **digitized head-shape points** (Polhemus) in the *head* coordinate system.
- used for **spherical harmonic expansion** of the magnetic field during Maxwell filtering.
- must be stable across runs if runs are to be combined.

These two products — **pose** and **center** — serve fundamentally different mathematical roles in SSS/tSSS and must not be mixed up.

Your pipeline separates the two clearly:

- **cHPI pose** → **head_pos** array or **.pos** file
- **head center** → **sphere origin fitted from digitization**

Both are required for robust, reproducible Maxwell filtering.

2. Why Head Position Matters in MEG

Magnetometers and gradiometers measure the magnetic field in a fixed coordinate frame attached to the dewar.

The subject, however, **moves inside the helmet**.

MEGIN's continuous HPI system embeds five (or more) small coils oscillating at fixed frequencies (≈ 293 – 321 Hz). These act as *fiducials* in continuous time. MNE:

1. Extracts each coil's amplitude and phase.
2. Solves a nonlinear localization problem to track each coil in 3D.
3. Computes the best-fit rigid-body transform to map coil positions into a **6-DOF pose** (x, y, z, yaw, pitch, roll).
4. Produces a **t × 7** matrix

```
[ time, tx, ty, tz, rx, ry, rz ]
```

This matrix is what SSS uses for **movement compensation**.

Meanwhile, the **SSS origin** determines the *mathematical center of the spherical model* used to expand internal and external field components. It is **not** derived from cHPI. It comes from Polhemus digitization.

3. Coordinate Frames: Device, Head, and Meaning of Each

MNE must move cleanly between:

- **Device frame**: fixed to the MEG helmet (SQUIDs).
- **Head frame**: fixed to the subject, defined by nasion + LPA + RPA digitization.
- **Coil frame**: location of each cHPI coil relative to the head.

The crucial facts:

Quantity	Frame	Source	Purpose
Head center / SSS origin	HEAD	Polhemus digitization	Defines the sphere for SSS math
cHPI coil positions	DEVICE	MEG sensors	Tracks movement
Rigid transform (DEVICE→HEAD)	Both	Derived from fiducials	Aligns spaces
head_pos time series	DEVICE? (time)	cHPI	Movement compensation only

If the **origin** is wrong → SSS is mathematically incorrect, leading to:

- distorted fields
- inflated amplitudes
- failure of shielding separation
- downstream bad epoch classification (as you observed)

If the **pose** is wrong or misaligned with the cropped time window →

- tSSS fails to apply movement correction properly
- large distortions occur in gradiometers
- autoreject sees the data as nonsense

Your refactor makes pose and origin independent and explicit.

4. The Two Products You Must Get Right

4.1 Head-Movement Time Series (head_pos)

This is the **array or .pos file**:

```
time, tx, ty, tz, rx, ry, rz
```

- **time is in absolute seconds**, usually matching `raw.first_time`.
- Rotations are in radians.
- Must be clipped to the same window as the Raw object.
- If cropped incorrectly, movement compensation in SSS becomes garbage.

4.2 Head Center ("SSS Origin")

Derived from digitized head-shape points:

- Polhemus captures EXTRA + CARDINAL points in the HEAD frame.
- `fit_sphere_to_headshape(info)` returns a sphere center in **HEAD** coordinates.
- This is fed to `origin=` in `maxwell_filter`.
- Should **not vary across runs** for a given subject/session.

Your old `apply_maxwell_filter()` used this correctly.

5. Why Sub-windowed Data Is Hard

When you analyze **only a 10-minute window** from a 60-minute recording:

- `raw.first_time` shifts (e.g., from 0 to ~643.5 s).
- The `.pos` file still spans 0–3600 s.
- So your pipeline must **clip and realign** the head-pos time axis.
- If this step is even slightly wrong → SSS sees mismatched times → fields explode → autoreject flags everything.

Your new `align_headpos_to_cropped_raw()` resolves this, provided the origin is still fit correctly.

6. Using Multiple Runs and Consistent Centers

If you want **Run 1, Run 2, ...** to be processed in a comparable way:

- Use the **same SSS origin** for all runs (usually median or fitted origin).
- Optionally use the **same destination coordinate** (e.g., median pose of Run 1).
- cHPI poses may be taken from different runs (`source="run"` in YAML) if needed.

Consistent origins → better between-run alignment

Consistent destinations → stable transforms for movement compensation

Consistent clipping → no timing mismatches

7. YAML Parameter Reference

Below is the updated block used in your pipeline.

```
head_position_processing:
```

```

# -----
# HOW TO GET HEAD-POSE DATA
# -----
source: "compute"          # "compute" | "file" | "coordinates" |
"run" | "none"

# If source="file"
file_path: "/path/to/headpos.pos"

# If source="coordinates" (no movement comp)
coordinates: [-0.075, 0.012, 0.030]

# If source="run" (borrow pose from run N)
reference_run: 1

# -----
# HOW TO CHOOSE DESTINATION
# -----
destination: "median"      # "median" | "mean" | "first" | "last" |
"coordinates" | "reference" | "none"

# destination="coordinates"
destination_coordinates: [-0.075, 0.012, 0.030]

# destination="reference"
destination_reference_run: 1

# -----
# MOVEMENT COMPENSATION
# -----
movement_compensation: true # true = use head_pos; false = static
maxwell

# -----
# ADVANCED OPTIONS
# -----
force_recompute: false     # ignore existing .pos file
write_subset: true         # write clipped, windowed .pos file
subset_naming: "desc-crop" # filename tag for cropped .pos

```

8. Parameter Interpretation and When to Use Each

8.1 `source: compute`

Compute cHPI pose **directly from the current raw segment**.

Best when:

- you want movement compensation tailored to the analysis window
- no .pos file exists
- the cHPI coils are stable

8.2 source: file

Load an existing `.pos`.

Best when:

- you already computed pose for the *full run*
- you want reproducible analysis across subsets
- computing pose is very slow (1+ hour for 60 minutes)

8.3 source: run

Useful when:

- analyzing multiple runs
- you want them aligned to a common pose reference (Run 1)

8.4 source: coordinates

No movement compensation; just use a static origin.

Rare, but appropriate when:

- cHPI failed
- subject barely moved

8.5 destination options

Controls the **target pose** used when aligning SSS before spatial filtering.

- `"median"` → best all-purpose choice
- `"first"` → stable if early posture matters
- `"coordinates"` → for reproducible research across sessions
- `"none"` → disables movement compensation

8.6 movement_compensation

- `true` → use full tSSS + dynamic transforms
- `false` → SSS only, no movement paths

Turn this off only when cHPI data are bad.

8.7 write_subset

Writes a clipped `.pos` file matching the analysis window.

This is extremely helpful for debug and documentation.

9. Practical Workflow Examples

9.1 Full-run preprocessing (1 hour)

```
source: "compute"
destination: "median"
movement_compensation: true
```

9.2 Analyze only a 10-min window, but reuse full-run head pose

```
source: "file"
file_path: "..._headpos.pos"
destination: "median"
movement_compensation: true
```

9.3 Multi-run experiment with aligned centers

```
source: "run"
reference_run: 1
destination: "reference"
destination_reference_run: 1
movement_compensation: true
```

9.4 No cHPI available (rare)

```
source: "coordinates"
coordinates: [-0.07, 0.02, 0.03]
movement_compensation: false
```

10. Conceptual Summary

Concept	Description
cHPI-derived pose	Time-varying head movement; needed for dynamic SSS.
SSS origin (head center)	Static geometric model fitted from Polhemus points.
Destination pose	Where to “align” the head before harmonic expansion.
Movement compensation	Corrects MEG data for within-run motion.
Cropping/head-pos realignment	Ensures pose and data use matching absolute time bases.
Multiple-run alignment	Ensures shared coordinate frame across experimental runs.

Key principle:

Head movement and head center are mathematically independent.

SSS requires both of them to be correct and in the right frame.

With correctly clipped head-pos arrays and consistent SSS origins, Maxwell filtering becomes stable, reproducible, and interpretable.